

## SOLUTION DOCUMENT

# Jeeran Network

A hyperlocal advertising network connecting neighborhood shops through a shared customer wallet pass — offers that follow the customer, and QR stands that turn each shop into a discovery point for the shops around it.

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Version 1.0 · August 2026

## 1. Executive Summary

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Jeeran Network lets small, local businesses advertise to each other's foot traffic instead of buying expensive, broad digital ads. A shop displays a Jeeran QR stand; when a customer scans it, they see live offers from *nearby, non-competing* businesses and can save a single persistent "Jeeran Offers" pass to their Google Wallet. From then on, that pass is pushed fresh nearby offers automatically — using the phone's own on-device location awareness, not a tracking service Jeeran operates — every time the customer is near a participating shop.

Shops earn advertising credit for recruiting customers and for repeat visits, spend credit to have their own offers surfaced, and climb a visible loyalty tier (Bronze → Silver → Gold → Platinum) that gives them a ranking edge among their peers. The whole loop — discovery, save, push, redeem, reward — runs without a companion app, without paid ad placement, and without the shop ever seeing another shop's customer data.

### For customers

One wallet pass, no app install, offers that arrive on their own as they move around the neighborhood.

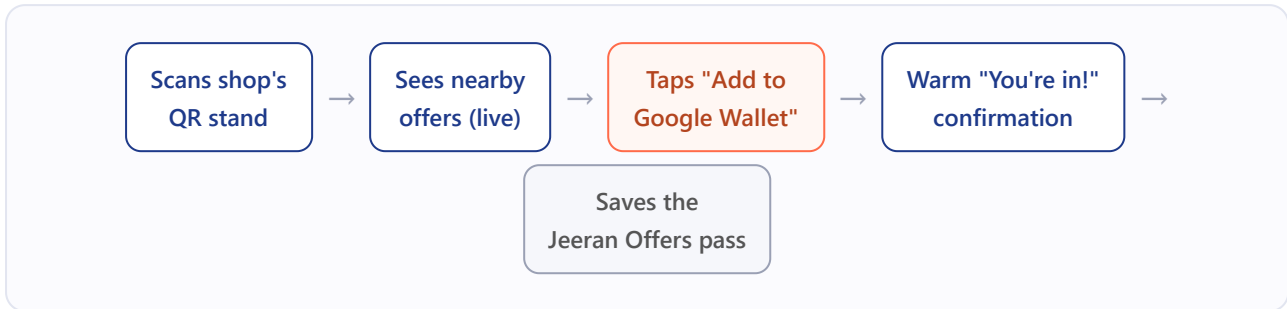
### For shops

Free customer acquisition from neighbors' foot traffic, a bid-based (not pay-to-play) placement model, and a loyalty program that rewards genuine activity.

### For the platform

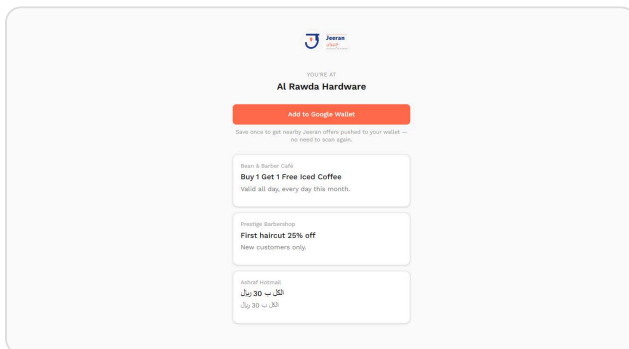
A self-reinforcing network effect — every new shop is both an advertiser and a new discovery point for every other shop.

## 2. How It Works — Customer Journey

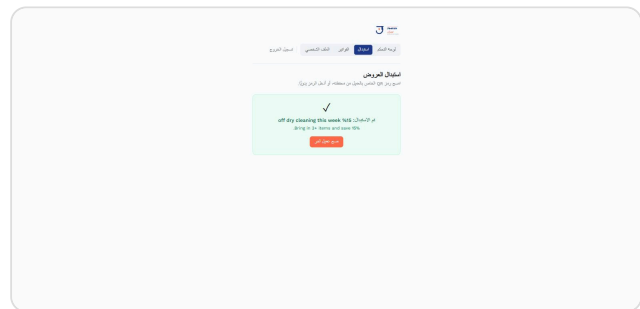


The pass is **persistent, not per-offer** — one save covers every future offer the customer will ever see through Jeeran. Once saved:

- The pass's card content refreshes automatically whenever a nearby offer changes (a shop activates, edits, or deactivates a campaign) or on a daily safety-net sweep.
- The pass carries up to 10 real shop coordinates in Google Wallet's own `merchantLocations` field — the OS itself compares this against the phone's live GPS and can surface the pass on the lock screen near any of them, entirely on-device. Jeeran's server never sees or stores the customer's live position.
- A lock-screen notification fires for a genuinely new offer nearby (not for every reshuffle), respecting Google's own 3-per-24h cap per pass.
- The card and its fixed labels render in the customer's own device language automatically (Arabic, English, or Urdu) — Google Wallet resolves this client-side from a translation table Jeeran provides, with no need to know or store each customer's language.

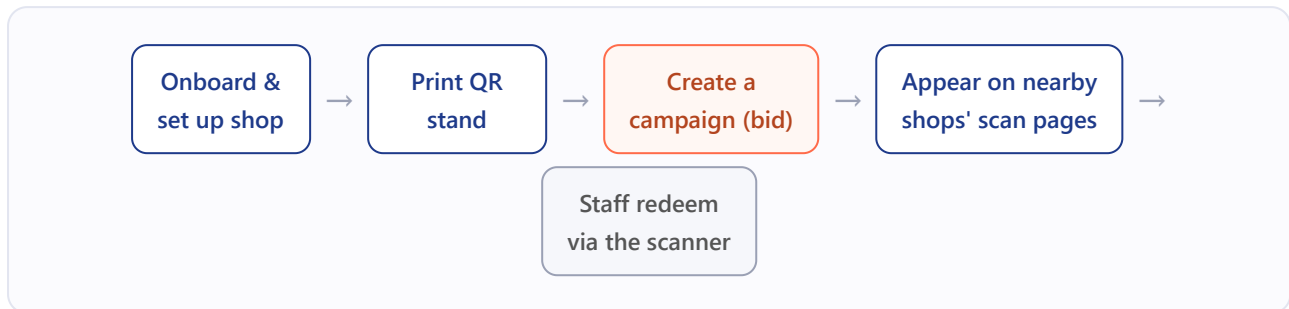


Scan-landing page — live nearby offers



Staff-side redemption confirmation

### 3. How It Works — Shop Journey



A shop's own campaign never appears on its own scan page, and never appears on a same-category competitor's page (a second salon never sees another salon's offer). Placement among the eligible competitors is decided by an auction: highest **bid\_per\_view** wins, with the shop's loyalty tier breaking ties when bids are close — never overriding a real bid gap.

#### Earning & spending credit

| Event                                           | Who earns                                                       | Notes                                                                                                                               |
|-------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| New customer recruited (first scan ever)        | Recruiting shop                                                 | One-time flat bonus, daily per-shop cap                                                                                             |
| Existing customer rescans a QR stand            | Scanned shop <i>and</i> the customer's original recruiting shop | Two independent line items; both trigger together if it's the same shop. 24h cooldown per (customer, shop) pair, daily per-shop cap |
| Customer redeems an offer in-store              | The shop whose customer it originally was (permanent)           | Amount = the redeemed offer's own bid                                                                                               |
| Shop crosses into a new loyalty tier            | That shop                                                       | One-time milestone bonus, admin-configurable per tier, announced on the dashboard — not a silent balance change                     |
| Shop's own campaign is shown / wins a placement | — (spends, doesn't earn)                                        | Bid is only actually charged on redemption, not on view                                                                             |

## 4. Loyalty Tiers

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Every shop's tier is computed live from its total redemptions earned as an advertiser — never stored, always recalculated. Thresholds are admin-configurable from one shared settings screen, not hardcoded, and every ranking function that reads tier reads the same table.

### **Bronze**

The floor everyone starts at.

### **Silver**

Default threshold: 5 redemptions.

### **Gold**

Default threshold: 20 redemptions.

### **Platinum**

Default threshold: 50 redemptions.

### **Milestone bonus**

One-time credit the moment a shop first crosses into a tier — announced with a dismissible dashboard banner, not a quiet number change.

### **Ranking effect**

Tier is a *tiebreaker*, not an override — it only decides ordering when two shops' bids are already close.

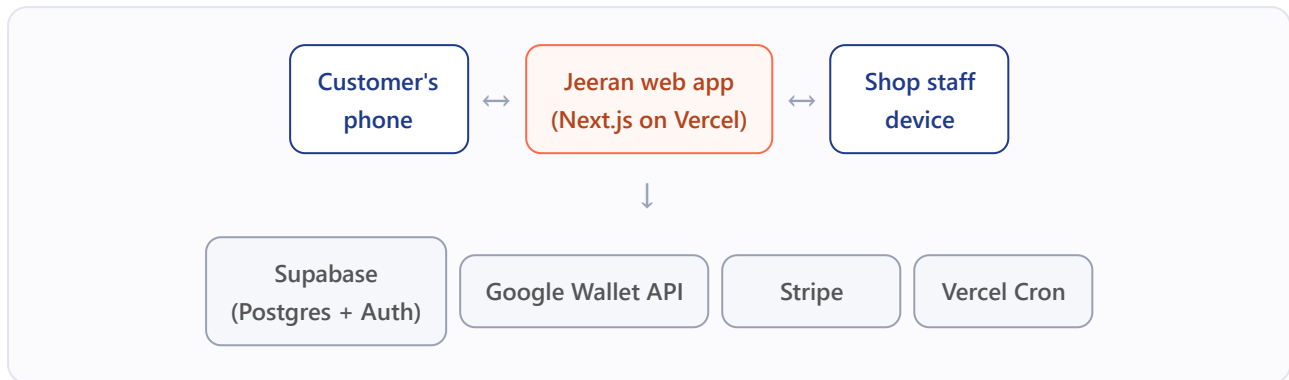
Shops also see a "Most Active Shops" leaderboard (this week / this month), ranked by genuine new-customer acquisition only — rescans never count. Purely informational; nothing about it feeds back into ranking or credit.

## 5. Feature Inventory

| Area                  | Feature                           | Summary                                                                                                                |
|-----------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Wallet pass           | Persistent membership pass        | One Google Wallet object per customer, updated in place rather than one pass per offer                                 |
|                       | On-device geofencing              | <code>merchantLocations</code> field — up to 10 nearby shop points, resolved entirely by the OS                        |
|                       | Localized card content            | Fixed labels shown in the viewer's own device language (ar / en / ur)                                                  |
|                       | Lock-screen notifications         | Fires only for genuinely new offers, via Google's typed <code>addMessage</code> / <code>TEXT_AND_NOTIFY</code> API     |
| Matchmaking           | Scan-page auction                 | Highest bid wins, excludes host, same-category competitors, and linked technicians                                     |
|                       | Geo-push ranking                  | Bid primary, tier breaks close-bid ties, distance is the final tiebreak                                                |
|                       | Event-driven + safety-net refresh | Pushes fire on campaign activate/deactivate/edit; a daily sweep catches everything else (lapsed dates, expired trials) |
| Redemption            | Staff scanner                     | Camera scan (native API + a JS fallback for Safari/iOS) plus manual entry, always available                            |
|                       | Opaque signed barcode             | HMAC-signed token, not a URL — safe if scanned by a generic camera app                                                 |
|                       | Distinct, localized outcomes      | Staff see exactly why a scan failed — no active offer, already redeemed, dead pass, bad code                           |
| Rewards & attribution | Permanent first-touch attribution | The first shop that ever recruited a customer keeps redemption credit for that customer's lifetime                     |
|                       | Rescan-credit mechanic            | Dual award, cooldown- and cap-guarded, IP-logged for fraud review                                                      |
|                       | Signup bonus                      | One-time, capped per shop per day                                                                                      |
|                       | Milestone bonus                   | One-time per tier crossing, admin-configurable amount, announced                                                       |
| Platform              | Admin settings                    | Tier thresholds, bid-tiebreak range, and milestone amounts — one shared, editable source                               |
|                       | Billing                           | Stripe subscriptions + credit top-ups, 3-month trial lifecycle with warning + auto-lapse                               |

## 6. Architecture at a Glance

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See the *High-Level Design* document for the full component breakdown, and the *Low-Level Design* document for schema, function, and API detail.

## 7. Security & Data Handling

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- **No live location tracking.** Jeeran never receives or stores a customer's ongoing position — the "follows you" behavior is delivered entirely on-device by the Wallet app comparing the phone's own GPS against points Jeeran already pushed.
- **Opaque, signed identifiers.** The pass barcode and the legacy one-off coupon links are HMAC-signed server-side secrets, not guessable or forgeable.
- **Deny-by-default data access.** Customer wallet data ( `wallet_members` , touchpoints, credit events) is only ever read or written by the server's own service role — never exposed to a browser session directly.
- **Anti-farming guards, with an accepted residual risk.** Daily per-shop caps and 24h cooldowns bound how much a single device or a small coordinated group can extract from the reward mechanics; IP is logged on signups and rescans for manual fraud review, not as an automatic block. This does not eliminate a determined small-scale coordinated abuse pattern — that trade-off is deliberate, matching the effort level appropriate for the platform's current stage.
- **Known, tracked gap:** the internal admin panel has no authentication yet. It is intentionally not linked from any public page and is flagged for lock-down before wider exposure.



## 8. Current Limitations & Roadmap

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| Area                                   | Current state                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apple Wallet                           | The one-off coupon flow supports Apple; the persistent membership pass is Google-only today. Apple's equivalent <code>relevantLocations</code> /geofencing mechanism is documented for whoever builds that path next. |
| Offer content translation              | Fixed pass labels are localized; individual offer titles/descriptions are shown exactly as the shop typed them, in whichever language that was.                                                                       |
| Admin authentication                   | Not yet built — a known, tracked gap, not an oversight.                                                                                                                                                               |
| Tier progress view & discovery counter | Scoped but intentionally held back pending the tier-ranking work landing first (now complete).                                                                                                                        |